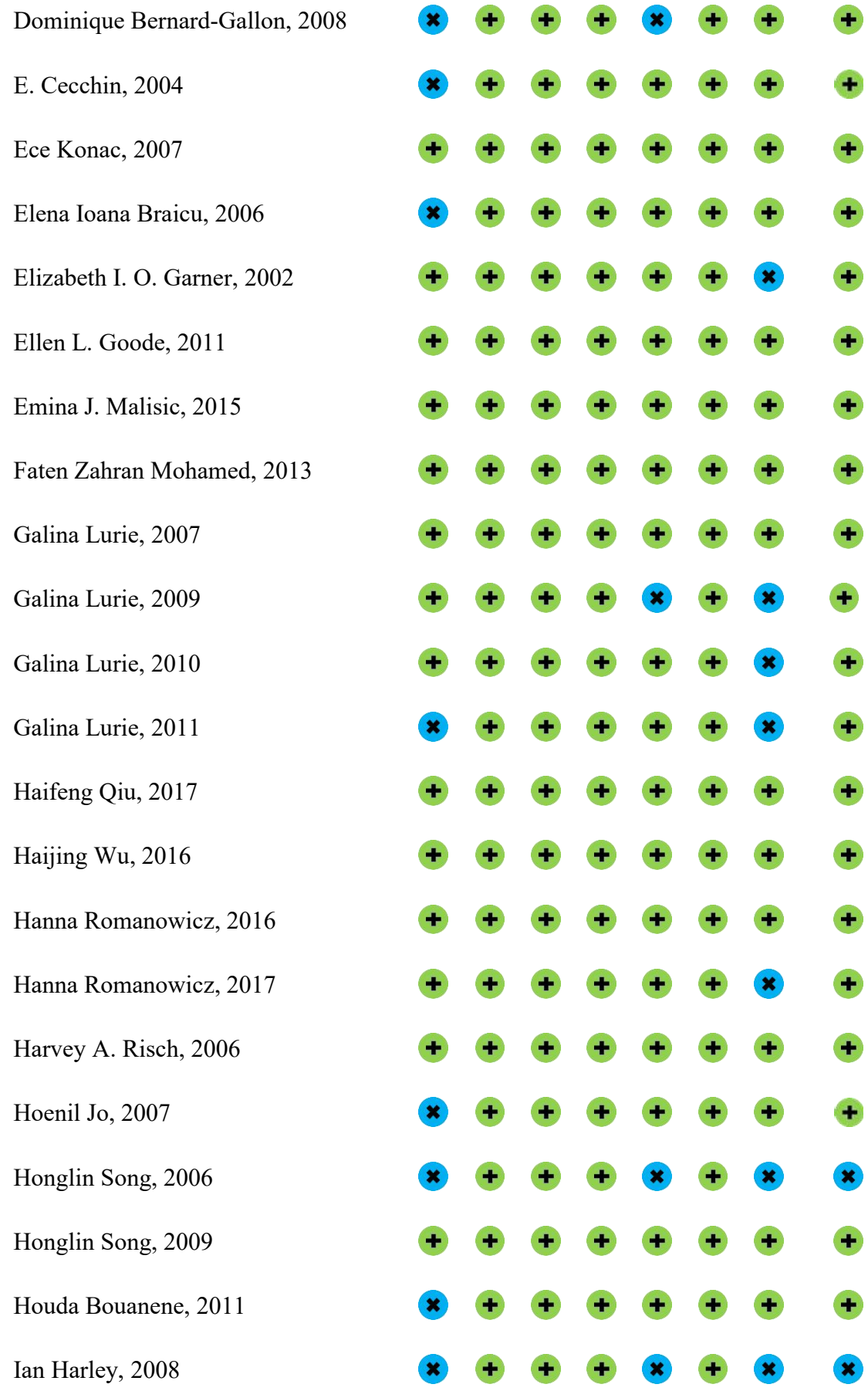
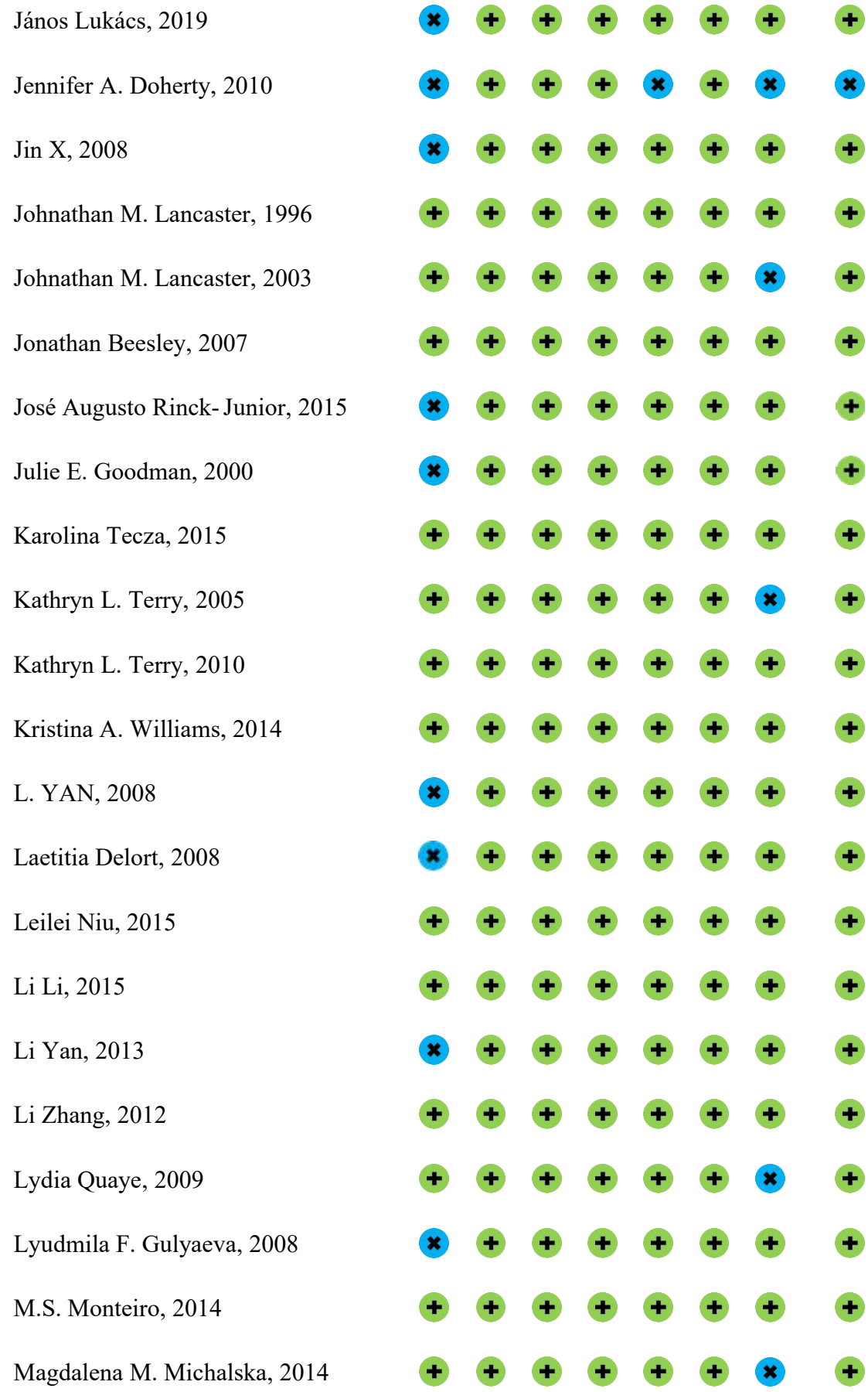
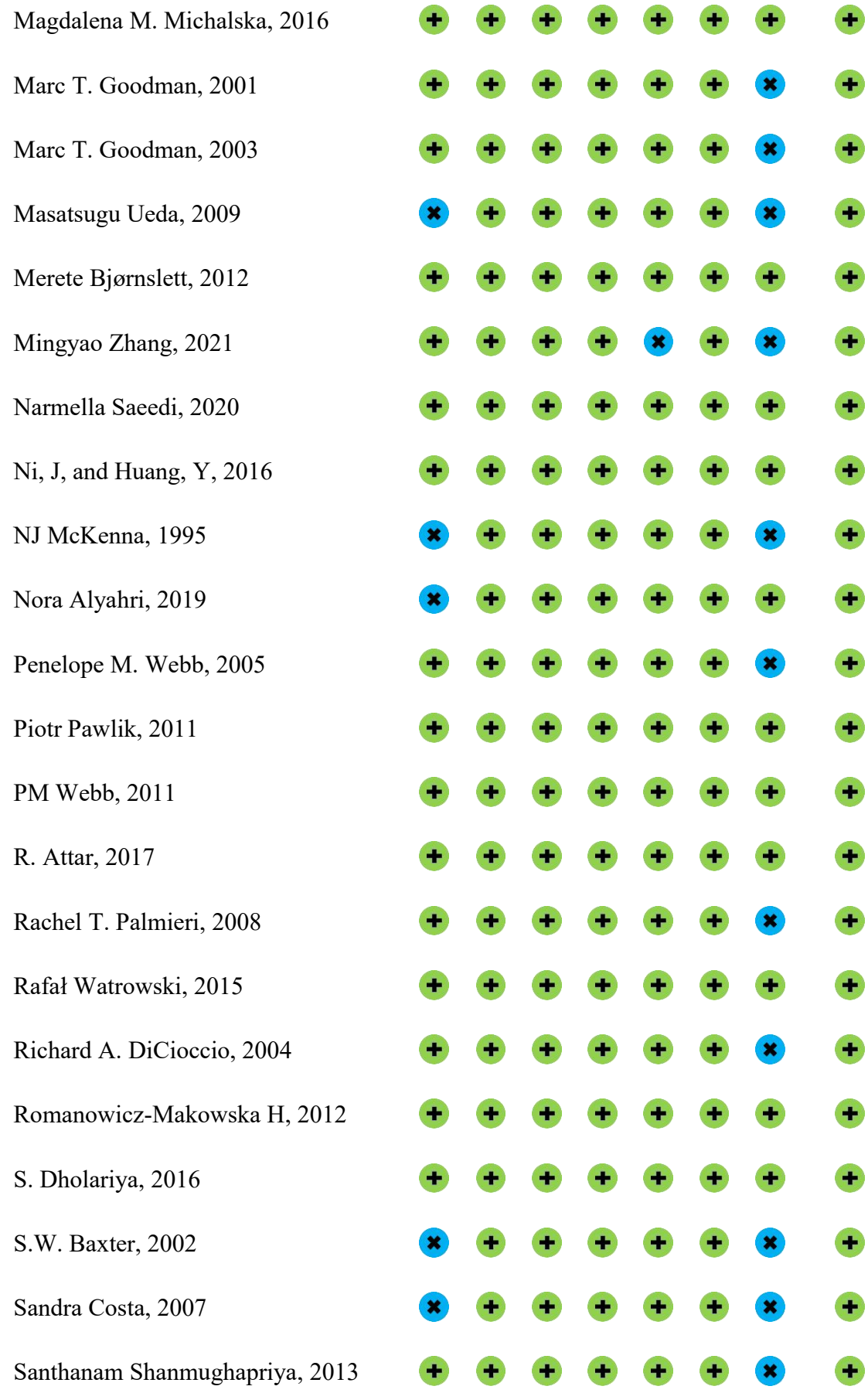
Supplement information 4. Risk of bias (ROBINS-1) assessment for included studies.

	①	②	③	④	⑤	⑥	⑦	Overall
Abigail W. Bushley, 2004	+	+	+	+	×	+	×	+
Adrianna Mostowska, 2013	+	+	+	+	+	+	+	+
Adrianna Mostowska, 2016	+	+	+	+	+	+	+	+
Agnieszka Honorata Ludwig, 2009	×	+	+	+	+	+	×	+
Alamtaj Samsami Dehaghani, 2009	×	+	+	+	+	+	+	+
Amal M.H. Mackawy, 2019	×	+	+	+	+	+	×	+
Amanda B.Spurdle, 2001	+	+	+	+	+	+	×	+
Andrea Romano, 2006	+	+	+	+	+	+	×	+
Andrew Berchuck, 2004	+	+	+	+	+	+	+	+
Anıl Çağla Özkılıç, 2016	×	+	+	+	+	+	+	+
Anna Jakubowska, 2007	+	+	+	+	+	+	×	+
Anna Jakubowska, 2010	×	+	+	+	+	+	+	+
Annika Auranen, 2005	+	+	+	+	×	+	×	+
BEATA SMOLARZ, 2013	+	+	+	+	+	+	+	+
Beata Smolarz, 2019	×	+	+	+	+	+	+	+
Catherine M. Phelan, 2010	+	+	+	+	×	+	×	+
D. V. Khokhrin, 2012	×	+	+	+	×	+	×	×
Dan Tong, 2001	+	+	+	+	+	+	+	+
Daniela B. Leite, 2008	×	+	+	+	+	+	×	+
Delores J. Grant, 2013	+	+	+	+	+	+	+	+


Dominique Bernard-Gallon, 2008
E. Cecchin, 2004
Ece Konac, 2007
Elena Ioana Braicu, 2006
Elizabeth I. O. Garner, 2002
Ellen L. Goode, 2011
Emina J. Malisic, 2015
Faten Zahran Mohamed, 2013
Galina Lurie, 2007
Galina Lurie, 2009
Galina Lurie, 2010
Galina Lurie, 2011
Haifeng Qiu, 2017
Haijing Wu, 2016
Hanna Romanowicz, 2016
Hanna Romanowicz, 2017
Harvey A. Risch, 2006
Hoenil Jo, 2007
Honglin Song, 2006
Honglin Song, 2009
Houda Bouanene, 2011
Ian Harley, 2008


János Lukács, 2019
Jennifer A. Doherty, 2010
Jin X, 2008
Johnathan M. Lancaster, 1996
Johnathan M. Lancaster, 2003
Jonathan Beesley, 2007
José Augusto Rinck- Junior, 2015
Julie E. Goodman, 2000
Karolina Tecza, 2015
Kathryn L. Terry, 2005
Kathryn L. Terry, 2010
Kristina A. Williams, 2014
L. YAN, 2008
Laetitia Delort, 2008
Leilei Niu, 2015
Li Li, 2015
Li Yan, 2013
Li Zhang, 2012
Lydia Quaye, 2009
Lyudmila F. Gulyaeva, 2008
M.S. Monteiro, 2014
Magdalena M. Michalska, 2014


Magdalena M. Michalska, 2016
Marc T. Goodman, 2001
Marc T. Goodman, 2003
Masatsugu Ueda, 2009
Merete Bjørnslett, 2012
Mingyao Zhang, 2021
Narmella Saeedi, 2020
Ni, J, and Huang, Y, 2016
NJ McKenna, 1995
Nora Alyahri, 2019
Penelope M. Webb, 2005
Piotr Pawlik, 2011
PM Webb, 2011
R. Attar, 2017
Rachel T. Palmieri, 2008
Rafał Watrowski, 2015
Richard A. DiCioccio, 2004
Romanowicz-Makowska H, 2012
S. Dholariya, 2016
S.W. Baxter, 2002
Sandra Costa, 2007
Santhanam Shanmughapriya, 2013

Sarah K. Holt, 2007
Shan Kang, 2004
Shan Kang, 2008
Shan-Yang He, 2012
Shelley S. Tworoger, 2009
Simon A. Gayther, 2007
Simone P. Pinheiro, 2010
Song CX, 2012
Song Gao, 2012
Stian Knappskog, 2011
Susanne Schüler, 2014
Tadeusz Debniak, 2006
Tess V. Clendenen, 2008
Thomas A. Sellers, 2005
Thomas A. Sellers, 2008
TP Manolitsas, 1997
UZAY GORMUS, 2007
Vidudala V.T.S. Prasad, 2011
Wendy M. Smith, 2001
X.C. Sun, 2016
Xiaohong Zhang, 2013
Xiaoyan Liu, 2015

Judgement: Low Moderate

Note: ①Bias due to confounding; ②Bias due to selection of participants; ③Bias in classification of interventions; ④Bias due to deviations from intended interventions; ⑤Bias due to missing data; ⑥Bias in measurement of outcomes; ⑦Bias in selection of the reported result.